

**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL COAST REGION
81 Higuera Street
San Luis Obispo, California 93401**

ORDER NO. 97-32

**WASTE DISCHARGE AND WATER
RECLAMATION REQUIREMENTS
FOR
CAÑADA WOODS RECLAMATION COMPANY
(Producer and User)
MONTEREY COUNTY**

The California Regional Water Quality Control Board, Central Coast Region, (hereafter the Board) finds:

1. Clint Eastwood and Margaret Eastwood, owners of the Cañada Woods Reclamation Company and the Cañada Woods Golf Course, submitted a complete Report of Waste Discharge on March 7, 1997 in accordance with Section 13260 of the California Water Code. The report was filed for authorization to discharge treated domestic wastewater within the Monterey Peninsula and the Carmel River Hydrologic Areas.
2. The Cañada Woods Reclamation Company (hereafter Discharger, Producer) and the Cañada Woods Golf Course (hereafter User), proposes construction and operation of a wastewater treatment and disposal system for a residential development project located in Monterey County along Highway 68 and Carmel Valley. The wastewater treatment facility will serve flows from Cañada Woods Subdivision with 56 residential lots, 4 commercial units, 15 employee units, the Cañada Woods North Subdivision with 34 residential lots, a golf course, club house, recreation and equestrian facilities, and Monterra Ranch Subdivision with 42 low income and 171 residential units. The three residential subdivisions and the proposed treatment and disposal facility (Section 1, T16S, R1E, MD B&M Seaside quad.); are shown on attachment "A" of this order.
3. Average dry weather flows from the three subdivisions is estimated at 100,000 gallons per day. Flow estimates are based on per household water consumption rates for existing facilities in

the area. Flow estimates include a 10% contingency factor. An alternative sewer design will utilize septic tanks at each lot for solids removal and small diameter effluent lines for transmission of partially treated effluent to a central reclamation facility. The wastewater reclamation facility will use sequencing batch reactors with coagulation, flocculation, and disinfection for the production of tertiary treated water. After tertiary treatment, reclaimed water will be delivered or stored for reuse. Golf course spray irrigation is the primary method of disposal.

4. Individual septic tanks, and pumps will be maintained by each property owner, except in the Monterra Ranch inclusionary housing area where septic tanks and piping will be owned and operated under a County Service Area. The wastewater treatment and disposal system will be owned and operated by the Carmel Reclamation Company, a public utility company. The Discharger will provide water and sewer service to the three subdivisions. In the event of improper operation and maintenance of the reclamation facility, the County Public Works Department has agreed that the subdivisions may annex into Community Service Area No. 100 for back up service.
5. Short term emergency (e.g., power failure) of raw waste water storage will be provided by individual septic tanks, lift stations, and storage tanks. Total short term storage is estimated at approximately 373,000 gallons, or almost 4 days of storage. Long term (120 day) reclaimed water

storage reservoirs will be used when irrigation is not allowed.

6. Reclaimed water will be spray irrigated during unsaturated soil conditions at the Cañada Woods North golf course. The estimated 100 acre-feet of reclaimed water will supply 70 percent of the irrigation water demand at the 50 acre golf course. The balance of needed irrigation water will be from wells located on the property. Total nitrogen in the reclamation discharge will be taken up by specifically selected salt resistant turf grasses, resulting in minimal pollutant discharge to groundwater. During saturated soil conditions (i.e., no irrigation), reclaimed water will be directed to storage ponds for later use. A contractual agreement will be formalized to assure disposal in perpetuity of reclaimed tertiary treated water at the golf course.
7. Soils in the area consist of shale, clay/loam overlying fractured weathered shale bedrock. Groundwater occurs at depths from 25 to 40 feet beneath the ground surface.
8. The Water Quality Control Plan, Central Coastal Basin, (Basin Plan), was revised and adopted by the Board on September 8, 1994. The Basin Plan incorporates statewide plans and policies by reference and contains a strategy for protecting beneficial uses of State waters.
9. Beneficial uses of groundwater in the vicinity of areas reclaimed water use and disposal include Domestic, Agricultural, and Industrial Water Supply.
10. State Department of Health Services' criteria for use of reclaimed water is in Title 22, Chapter 3, of the California Code of Regulations. The Board has consulted with this Department regarding the regulation of this discharge.
11. The Supervisors of Monterey County certified a final Environmental Impact Report on December 17, 1996 in accordance with the California Environmental Quality Act (Public Resources Code, Section 21000, et seq.) and the California Code of Regulations.

CEQA requires a responsible agency to make certain findings regarding that portion of the project being approved by the responsible agency. Regarding the portion of the project that a responsible agency approves, it must mitigate or avoid any significant environmental effect that is within its powers. (CEQA Guidelines sec. 15096 (g).) The Regional Board's powers are limited to protection of water quality and abatement of nuisance associated with waste discharge. Therefore, this resolution includes findings regarding mitigation of the environmental effects identified in the EIR in connection with operation of the wastewater treatment plant and collection system and reclamation of treated effluent which involve water quality. The EIR does not identify any significant environmental impacts that cause a nuisance associated with waste discharge.

The EIR in Chapter 4.5 identifies the following significant environmental effect associated with operation of the wastewater treatment plant and collection and reclamation system which involves water quality:

"Impact #10: The proposed project would result in the generation of increased wastewater flows that can be accommodated by the proposed wastewater collection, treatment, and disposal facilities of wastewater. However, some adverse impacts to public health or safety could result if facilities are not properly maintained."

The EIR proposes the following mitigation measures to reduce this potential significant effect to insignificance:

"Mitigation

10-1 Design, construct and operate the proposed wastewater collection, treatment, and disposal facilities in accordance with all applicable state and county requirements, as planned, including but not limited to

- For individual residences a minimum 1,500 gallon septic tank should be installed to provide primary treatment, with tank sizes increasing by 500 gallons for each additional bedroom over and above three;
- Nitrate-nitrogen limit of 6 mg/l shall be required for the tertiary effluent;

- Short-term storage requirement of 300,000 gallons of raw wastewater shall be provided in strategically-located tanks within the collection system, with appropriate pumping and odor control facilities
- Long-term, wet weather storage requirements of 120 days of average flow, plus incident rainfall (approximately 45 AF total) shall be provided in lined storage ponds;
- Setback requirements from areas where reclaimed water is being spray irrigated include 25 feet from property line and 100 feet from streams and wells (no streams exist in the project vicinity);

10-2 Prohibit discharge of toxic substances or of substances into the wastewater system that would adversely affect the collection, treatment or disposal of the wastewater.

10-3 Operate the reclaimed water storage reservoir(s) to ensure the protection of public health and the environment, including implementation of the following measures:

- plant vegetation around the perimeter of the pond to act as a visual barrier and to limit public access;
- control algae by a combination of aeration, addition of non-toxic chemicals, and promotion of duck weed."

Changes or alterations have been required in, or incorporated into the project which avoid or substantially lessen this significant environmental effect by changing the project to include all of the mitigation measures proposed by the EIR and quoted above, as well as more stringent requirements as necessary. These changes are evidenced by the requirements of this waste discharge requirements order No. 97-32, the report of waste discharge and the vested tentative map approved by Monterey County.

CEQA also requires a responsible agency to adopt a reporting and monitoring program for the changes made to the project or conditions of project approval, to mitigate or avoid environmental effects. The program should ensure compliance during project implementation. (Public Resources Code sec. 21081.6.) A monitoring and reporting requirement is attached and incorporated into these waste discharge

requirements entitled Monitoring and Reporting Program No. 97-32. Additionally Regional Board staff will make periodic site visits to assure facilities are constructed and operated to implement the mitigation measure required by these waste discharge requirements.

12. Discharge of waste is a privilege, not a right, and authorization to discharge is conditional upon the discharge complying with provisions of Division 7 of the California Water Code and any more stringent effluent limitations necessary to implement water quality control plans, to protect beneficial uses, and to prevent nuisance. Compliance with this Order should assure this and mitigate any potential adverse changes in water quality due to the discharge.

13. On April 17, 1997, the Board notified the Discharger and interested persons of its intent to issue waste discharge requirements, provided them with an opportunity to submit their written views and recommendations, and scheduled a public hearing.

14. In a public hearing on July 11, 1997, the Board heard and considered all comments pertaining to the discharge and found this Order consistent with the above findings.

IT IS HEREBY ORDERED, pursuant to authority in Section 13263 of the California Water Code, that the Cafada Woods Reclamation Company, its agents, successors, and assigns, may spray dispose reclaimed wastewater to the Cafada Woods North golf course and grassland restoration area, providing they comply with the following:

(General permit conditions, definitions and the method of determining compliance are contained in the attached "Standard Provisions and Reporting Requirements for Waste Discharge Requirements," dated January, 1984. Applicable paragraphs are referenced in paragraph F.2. of this Order.)

Throughout the Order superscripts are provided to indicate the source of requirements specified. The superscripts are defined below.

A- Basin Plan

B - Wastewater Reclamation Criteria (Title 22, Division 4, Chapter 3. California Code of Regulations, dated 1978)

C - Mitigation measures in the Final EIR.

Requirements not referenced are based on staff's professional judgment for the sufficient protection of beneficial uses and water quality objectives.

A. PROHIBITIONS

1. Discharge to areas other than the Cañada Woods Golf Course, as shown in "Attachment A", is prohibited.
2. Discharge of any wastes including overflow, bypass, overspray, and seepage from septic tanks, collection system, treatment facility, or disposal areas to adjacent drainage ways and adjacent properties is prohibited.
3. Daily flow shall not exceed 100,000 gallons.
4. Discharge of self regenerating water softening brine to the wastewater collection system is prohibited^C.
5. The introduction of any substance into the waste stream which adversely affects the collection, treatment, or disposal facility is prohibited^C.
6. Storage pond freeboard shall not be less than 2 feet.
7. Discharge of reclaimed water not meeting the disinfected tertiary non-restricted reuse specifications to non-restricted reuse areas is prohibited.

B. SECONDARY 23 RESTRICTED RECLAMATION SPECIFICATIONS (MINIMUM SPECIFICATIONS)

1. All reclaimed water produced shall not have a pH less than 6.5 or greater than 8.4^A.
2. All reclaimed water produced shall be in conformance with the reclamation criteria established in Title 22, Division 4, Chapter 3, of the California Code of Regulations^B.

3. All reclaimed water produced shall at all times be adequately disinfected, oxidized wastewater^B.
4. All reclaimed water produced shall not contain constituents in excess of the following^B:

Parameter	Maximum
Total Dissolved Solids	1000 mg/L
Sodium	250 mg/L
Chloride	250 mg/L
Nitrate (as Nitrogen) ^C	6 mg/L

5. The median number of coliform organisms in all reclaimed water produced shall not exceed 23 per 100 milliliters, as determined from the bacteriological results of the last 7 days for which analyses have been completed, and the number of coliform organisms shall not exceed an MPN of 240 per 100 milliliters in any two consecutive samples^B.

C. DISINFECTED TERTIARY NON-RESTRICTED REUSE SPECIFICATIONS

1. Reclaimed water produced for non-restricted spray irrigation shall be at all times adequately disinfected, oxidized, coagulated, clarified, filtered wastewater^B.
2. Reclaimed water produced for non-restricted spray irrigation shall not contain constituents in excess of the following^B:

Constituent	Unit	30-day	Daily
		Average	Max.
BOD ₅	mg/l	5*	10
Turbidity*	NTU	2	5
Suspended Solids	mg/l	5*	10
Settleable Solids	ml/l	-	0.1

*Mean limit is the average of the highest hourly reading for any 24-hour period. Maximum limit shall not be exceeded more than 5 percent of the time during any 24-hour period. Standard Provisions require increased sampling frequency to determine compliance (Refer to General Monitoring Requirement B.2).

3. The median number of coliform organisms in reclaimed water produced for non-restricted spray irrigation shall not exceed 2.2 per 100 milliliters, as

determined from the bacteriological results of the last 7 days for which analyses have been completed, and the number of coliform organisms shall not exceed an MPN of 23 per 100 milliliters in more than one sample in any thirty (30) day period^B.

4. The Discharger shall provide a chlorine disinfection process that provides a CT (chlorine concentration time modal contact time) value of not less than 450 milligrams-minutes per liter at all times with a modal contact time of at least 90 minutes, based on peak dry weather design flow.

D. GENERAL REUSE SPECIFICATIONS

1. Use of reclaimed water shall be in conformance with the reclamation criteria established in Title 22, Division 4, Chapter 3, of the California Code of Regulations.
2. Drinking water fountains shall be protected from direct or windblown reclaimed water spray.
3. Reclaimed water shall be applied at a rate and volume as not to exceed vegetative demand and soil moisture conditions. Special precautions must be taken to prevent clogging of spray nozzles, to prevent overwatering and to minimize the production of runoff. Pipelines shall be maintained so as to prevent leaks.
4. Reclaimed water shall not be used during periods of extended rainfall and/or runoff.
5. Signs shall be provided to inform the public that reclaimed water is being stored or used.
6. Personnel involved in transporting or using reclaimed water shall have undergone adequate training in reclaimed water use, as provided by the User.
7. Reclaimed water piping (excluding existing landscape piping), controllers, valves, etc., shall be marked to differentiate reclaimed water facilities from potable water facilities.
8. Reclaimed water valves, outlets, quick couplers, and sprinklers shall be of a type, or secured in a

manner, that permits operation only by authorized personnel.

9. Use or installation of hose bibs on the reclaimed water system shall not be permitted.
10. Proper backflow and cross-connection protection for domestic water services and irrigation wells shall be provided.
11. Reclaimed water shall not be irrigated or impounded within 100 feet of any well used for domestic supply.
12. Reclaimed water systems shall be regularly inspected to assure proper operation, absence of leaks, and absence of illegal connections.

E. GROUND WATER LIMITATIONS

1. The discharge shall not cause a significant increase of mineral constituent concentrations in underlying ground waters.
2. The discharge shall not cause concentrations of chemicals and radionuclides in groundwater to exceed limits set forth in Title 22, Chapter 15, Articles 4, 4.5, 5 and 5.5 of the California Code of Regulations.^A

F. PROVISIONS

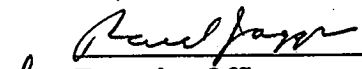
1. Discharger shall comply with "Monitoring and Reporting Program No. 97-32," as specified by the Executive Officer.
2. Discharger shall comply with all items of the attached "Standard Provisions and Reporting Requirements for Waste Discharge Requirements," dated January, 1984 (also referred to as "Standard Provisions"), except Item Nos. A.8, A.17, and C.16.
3. Pursuant to Title 23, Division 3, Chapter 9, of the California Code of Regulations, the Discharger must submit a written report to the Executive Officer not later than July 1, 2007, addressing:
 - a. Whether there will be changes in the continuity, character, location, or volume of the discharge; and,

July 11, 1997

- b. Whether, in their opinion, there is any portion of the Order that is incorrect, obsolete, or otherwise in need of revision.
4. The Discharger is required to restrict reclaimed water application to periods when the public is not present if monitoring shows non-compliance with disinfected tertiary non-restricted reuse specifications.
5. Non-restricted reuse of reclaimed water is subject to written approval of the Executive Officer of the Regional Board and concurrence of the State Department of Health Services.
6. If, in the opinion of the Executive Officer, the Discharger fails to comply with the provisions of this Order, including compliance with the above limits, such failure to comply shall constitute adequate grounds to initiate action for administrative civil liability, pursuant to Section 13323, or request the Attorney General take appropriate enforcement action pursuant to Section 13350 of the California Water Code.
7. The Discharger shall investigate and provide a description of a temporary alternate area* for disposal and shall submit a technical report to the Executive Officer of the Regional Board if treatment plant performance data or monitoring reports indicate reclaimed water is unsuitable for restricted or nonrestricted golf course irrigation.
- *The Regional Board will comply with the California Environmental Quality Act, before approving any alternative disposal area.
8. Use of recycled water during construction shall comply with the "Guidelines for the Use of Reclaimed Water for Construction Purposes" dated June 10, 1988.

I, ROGER W. BRIGGS, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Coast Region, on July 11, 1997.

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for Executive Officer

**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL COAST REGION
895 Aerovista Place, Suite 101
San Luis Obispo, California 93401**

MONITORING AND REPORTING PROGRAM NO. 97-32

Revised January 6, 2004

**For
CANADA WOODS RECLAMATION COMPANY
MONTEREY COUNTY**

EFFLUENT MONITORING

Representative samples of tertiary effluent shall be collected and analyzed according to the following schedule:

Constituent	Units	Sample Type	Minimum Frequency of Sampling and Analysis
Daily Flow	gal/day	Metered	Continuous
Turbidity	NTU	Metered	Continuous
Total Coliform	MPN/100 mL	Grab	Daily
Total Dissolved Solids	mg/L	Grab	Weekly
Sodium	mg/L	Grab	Weekly
Chloride	mg/L	Grab	Weekly
BOD ₅	mg/L	Grab	Weekly
Suspended Solids	mg/L	Grab	Weekly
Settleable Solids	mL/L	Grab	Weekly
pH	units	Grab	Weekly
Total Nitrogen ¹	mg/L as N	Grab	Weekly
Nitrate ¹	mg/L as N	Grab	Weekly

STORAGE FACILITY/SPRAY IRRIGATION AREA INSPECTION

Wastewater storage and disposal areas shall be inspected weekly. Notes shall be kept of observations and shall be summarized in annual monitoring reports. In the event of impending storage pond overflow, the Executive Officer and the County Environmental Health Department shall be notified immediately.

SEPTIC TANK MAINTENANCE

Septic tanks shall be inspected and pumped as follows. The inspection is not required during the year it is pumped.

Parameter	Units	Type of Measurement	Minimum Inspection Frequency
Sludge depth and scum thickness in each compartment of each septic tank	Feet	Staff Gauge	Annually (by April of each year)
Distance between bottom of scum layer and bottom of outlet device	Inches	Staff Gauge	Annually (by April of each year)
Distance between top of sludge layer and bottom of outlet device	Inches	Staff Gauge	Annually (by April of each year)

¹ Samples may be taken at the outlet of the recycled water storage reservoir, immediately prior to the golf course irrigation system intake.

Septic tanks shall be pumped when any one of the following conditions exist, or may occur before the next inspection:

- a. The combined thickness of sludge and scum exceeds one-third of the tank depth of the first compartment; or,
- b. The scum layer is within three inches of the outlet device; or,
- c. The sludge layer is within eight inches of the outlet device.

The annual report shall contain information regarding any problems noted during the previous year associated with operation, maintenance, or repair of any residential system septic tank, pump, or pipe. In the event of impending septic tank overflow, the Executive Officer and the County Environmental Health Department shall be notified immediately.

REPORTING

Monitoring results shall be submitted quarterly by the 20th day of January, April, July, and October. The report shall include a summary of any operational or equipment problems, reasons for any violations of waste discharge requirements, and corrective actions taken to eliminate violations.

Ordered by:

Bradley E. Hageman
Executive Officer

Date: 1/6/04